

Practical 4

Practice Lab Assignment

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4.1.1. Pandas - series creation and manipulation

29.4%

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

Submit

1 import pandas as pd

2

3 # Take inputs from the user to create a list of numbers

4 numbers = list(map(int, input().split()))

5

6 # Create a Pandas series from the list of numbers

7 s = pd.Series(numbers)

8 # Grouping by even and odd numbers and calculating the mean

Average time 0.009 s 8.83 ms Maximum time 0.022 s 22.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 0.022 ms Debug

Expected output 1 2 3 4 5 6 7 8 9 10

Actual output 1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers: 0

Mean of even and odd numbers: 0

Odd: 5.0

Odd: 5.0

Even: 6.0

Even: 6.0

dtype: float64

dtype: float64

Terminal Test cases

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4.1.2. Dictionary to dataframe

41.3%

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

datafram...

Submit

1 import pandas as pd

2

3 # Provided dictionary of lists

4 data = {

5 'Name': ['Alice', 'Bob', 'Charlie'],

6 'Age': [25, 30, 35],

7 }

8

Average time 0.075 s 76.50 ms Maximum time 0.088 s 88.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 0.088 ms Debug

Expected output Original DataFrame: 0 Alice 25 1 Bob 30 2 Charlie 35 New name: Susan

Actual output Original DataFrame: 0 Alice 25 1 Bob 30 2 Charlie 35 New name: Susan

Terminal Test cases

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4.1.3. Student Information

49.5%

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...

Submit

1 import pandas as pd

2

3 # Read the text file into a DataFrame

4 file = input()

5 data = pd.read_csv(file, sep="s+", header=None, names=["Name", "Age", "Grade"])

6

7

Average time 0.039 s 39.00 ms Maximum time 0.055 s 55.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1 0.055 ms Debug

Expected output studentdata.txt

Actual output studentdata.txt

First five rows: 0 John 25 A 1 Alin 22 B 2 Emma 24 A

First five rows: 0 John 25 A 1 Alin 22 B 2 Emma 24 A

Terminal Test cases

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4.2.1. Month with the Highest Total Sales

07:55

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for

monthFor...

Submit

Debugger

```
1 """import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
```

Average time: 0.025 s, Maximum time: 0.055 s

1 out of 1 shown test case(s) passed, 2 out of 2 hidden test case(s) passed

Test case 1: 45 ms, Debug

Expected output: sales_data.csv, Actual output: sales_data.csv

Best month: 2025-01, Best month: 2025-01

Total sales: \$1210.00, Total sales: \$1210.00

Terminal, Test cases

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4.2.2. Best Selling Product

08:00

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

monthFor...

Submit

Debugger

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8 product_sales = df.groupby('Product')['Quantity'].sum()
```

Average time: 0.018 s, Maximum time: 0.036 s

1 out of 1 shown test case(s) passed, 2 out of 2 hidden test case(s) passed

Test case 1: 36 ms, Debug

Expected output: sales_data.csv, Actual output: sales_data.csv

Best selling product: Product A, Best selling product: Product A

Total quantity sold: 23, Total quantity sold: 23

Terminal, Test cases

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4.2.3. City that Sold the Most Products

04:22

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

monthFor...

Submit

Debugger

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8 city_sales = df.groupby('City')['Quantity'].sum()
```

Average time: 0.022 s, Maximum time: 0.043 s

1 out of 1 shown test case(s) passed, 2 out of 2 hidden test case(s) passed

Test case 1: 43 ms, Debug

Expected output: sales_data.csv, Actual output: sales_data.csv

City sold the most products: Los Angeles, City sold the most products: Los Angeles

Terminal, Test cases

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4.2.4. Most Frequently Sold Product Pairs09:23

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Explanation:
Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C

frequent...

```
1 import pandas as pd
2 from itertools import combinations
3 from collections import Counter
4
5 # Prompt user to input the file name
6 file_name = input()
7
8 # Read data from the specified CSV file
```

Average time0.017 sMaximum time0.034 s1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 134 msDebug

Expected outputActual output
sales_data.csvsales_data.csv
Product A and Product B: 2 timesProduct A and Product B: 2 times
Product A and Product C: 2 timesProduct A and Product C: 2 times

TerminalTest cases

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4.2.5. Titanic Dataset Analysis and Data Cleaning08:45

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

- Display the first 5 rows of the dataset.
- Display the last 5 rows of the dataset.
- Get the shape of the dataset (number of rows and columns).
- Get a summary of the dataset (using `.info()`).
- Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
- Check for missing values and display the count of missing values for each column.
- Fill missing values in the 'Age' column with the median age.
- Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
- Drop the 'Cabin' column due to many missing values.
- Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-------	----------

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 # 1. Display the first 5 rows of the dataset
7 print(data.head())
8 # 2. Display the last 5 rows of the dataset
```

Average time0.180 sMaximum time0.180 s1 out of 1 shown test case(s) passed

Test case 1180 msDebug

Expected outputActual output
PassengerId Survived Pclass Fare Cabin Embarked
0 0 1 3 5.12 0 0
1 0 1 3 5.12 0 0
2 0 1 3 5.12 0 0
3 0 1 3 5.12 0 0
4 0 1 3 5.12 0 0

TerminalTest cases

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4.2.6. Titanic Dataset Analysis and Data Cleaning - 211:47

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
- Convert the 'Sex' column to numeric values (male: 0, female: 1).
- One-hot encode the 'Embarked' column, dropping the first category.
- Get the mean age of passengers.
- Get the median fare of passengers.
- Get the number of passengers by class.
- Get the number of passengers by gender.
- Get the number of passengers by survival status.
- Calculate the survival rate of passengers.
- Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-------	----------

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7 # 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)
8 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
```

Average time0.087 sMaximum time0.087 s1 out of 1 shown test case(s) passed

Test case 187 msDebug

Expected outputActual output
29.69911764705882 29.69911764705882
14.4542 14.4542
3 491 3 491
1 216 1 216
2 184 2 184

TerminalTest cases

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4.2.7. Titanic Dataset Analysis and Data Cleaning - 302/37

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Explorer

titanicDat...

Submit

Debugger

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
8 data = pd.get_dummies(data, columns=['Embarked'],
```

Average time0.090 s90.00 msMaximum time0.090 s90.00 ms1 out of 1 shown test case(s) passed

Test case 160 msDebug

Expected outputActual output

PclassPclass

1...0.6296301...0.629630

2...0.4728262...0.472826

3...0.2423633...0.242363

Name: Survived, dtype: float64Name: Survived, dtype: float64

Embarked_SEmbarked_S

TerminalTest cases

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 402/37

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Explorer

titanicDat...

Submit

Debugger

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data = pd.get_dummies(data, columns=['Embarked'],
7 drop_first=True)
8 # 1. Get the number of survivors by gender
```

Average time0.076 s76.00 msMaximum time0.076 s76.00 ms1 out of 1 shown test case(s) passed

Test case 176 msDebug

Expected outputActual output

female... 233female... 233

male... 109male... 109

Name: Sex, dtype: int64Name: Sex, dtype: int64

male... 468male... 468

female... 81female... 81

Name: Sex, dtype: int64Name: Sex, dtype: int64

TerminalTest cases

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